1.H1 Symmetries

Lesson Introduction

Benchmark	MA.912.GR.2.4 Determine symmetries of reflection, symmetries of rotation, and symmetries of translation of a geometric figure.	Learning Targets	• I can determine symmetries of reflection. • I can determine symmetries of rotation. • I can determine symmetries of translation.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• What is symmetry? • What symmetries can be used to create a tessellation?
Lesson Narrative	Students observe and discuss how transformations can be used to show symmetry. Following a pattern, students create and identify translational, reflectional, and rotational symmetries. Connections are made between the number of lines of symmetry with the order of rotational symmetry a polygon has. Tessellations are created using a pattern chosen by the individual students.

Component Summary				
	1.H1.1 Warm-Up (~5 min.)	Notice and Wonder about M.C. Escher artwork *(SE p. 48)*	1.H1.3 Activity (20 min.)	Creating a tessellation *(SE p. 51)*
	1.H1.2 Exploration (20 min.)	Describing and creating translational, reflectional, and rotational symmetries *(SE p. 48)*	1.H1.4 Wrap-Up (~5 min.)	Classifying transformations *(SE p. 52)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Tessellation <u>Additional Terms:</u> • Symmetry	• 3 x 5 Note Cards • 8.5 x 11 Paper • Scissors • (Optional) Colored Pencils • (Optional) Tangrams or Polygon Shapes	• After presenting the definition of tessellation in 1.H1.2 Exploration, consider projecting images of other tessellations, especially famous ones by M.C. Escher.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may only see one possible type of transformation.
Students may confuse horizontal and vertical cuts.</td><td>This lesson should be taught after Lesson 7 for Honors courses.
Consider instilling “stops” in the lesson to check in with partner pairs and their understanding. Suggested stops are included.
Rotational symmetry is a new concept. Consider doing question 7 in 1.H1.2 Exploration as a whole group.
Consider allowing students to complete 1.H1.3 Activity for homework.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
In 1.H1.1 Warm-Up, students write things they wonder about two images of M.C. Escher’s artwork. This activity sets the stage for students to look at transformational symmetries and tessellations.</td><td>MA.K12.MTR.5.1: Patterns and Structure
Students will develop a tessellation in 1.H1.3 Activity. This image will emphasize the importance of following a pattern. Students will be spending time focusing on relevant details in their pattern and creating plans to logically produce an overall image. This activity helps students recognize patterns in the world around them and make connections to mathematical concepts.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Students will use repeated translations to develop a tessellation pattern in 1.H1.3 Activity. This pattern will continue across an entire page and should end up as a clearly artistic representation of the mathematical concepts presented throughout Unit 1. The process of cutting and taping the 3 x 5 note cards will allow students to have a great kinesthetic connection with the topic. Once the tessellation is being developed, the aesthetics of the pattern will begin to speak to students' visual learning needs. This emphasis can be amplified by the addition of color and implementation of other key elements from other courses.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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1.H1.1 Warm-Up: Artwork of M.C. Escher

Component Overview: *Students view two images of M.C. Escher's artwork, writing anything they notice or wonder. Then, students state the similarities and differences in the images.*

1.H1.1 Warm-Up: Artwork of M.C. Escher

M.C. Escher (1898-1972) was a Dutch artist whose work was inspired by mathematics. Though initially trained as an architect, he became one of the most famous and influential graphic artists in the world.

Below are two examples of Escher's work.

Example A	Example B
	

1. Describe any similarities and differences you notice about the two examples.

Similarities	Differences
Answers will vary. Both pieces have images that are repeated. In both pieces, the repeated images are rotated.	*Answers will vary. Example A is black and white, and Example B is in color. In Example B, all the geckos are the same size, but in Example A, the bats and angels are different sizes.*

Consider having students write down two things they notice about each of the examples of Escher's work. Then, give students a chance to turn and talk about what they initially noticed about the artwork. Students can then complete question 1.

1.H1.2 Exploration: Symmetry

Component Overview: *Students determine which of five images have symmetry and what transformations show that one part of the image aligns with the other part(s) of the image. Students identify tessellations that were created using translational, reflectional, and rotational symmetry. A table is completed with the number of lines of symmetry and the order of rotational symmetry for several polygons.*

1.H1.2 Exploration: Symmetry

The five images shown below have symmetry.

Image A	Image B
	
Image C	**Image D**
	

Students may struggle most with Image B and Image C. Avoid providing any hints prior to working through the remainder of this component. Revisit these, and any other images, before moving on to 1.H1.3 Activity.

1. Discuss with your partner why each image has symmetry and which transformations could be used to show that one part of the image aligns with the other. *Answers will vary. They are either reflections or rotations.*

2. Complete the table. Some images may align with more than one type of transformation.

Image	Transformations
A	☐*reflection across a line* ☐rotation about a point ☐translation
B	☐reflection across a line ☐*rotation about a point* ☐translation
C	☐reflection across a line ☐rotation about a point ☐*translation*
D	☐*reflection across a line* ☐*rotation about a point* ☐translation

1.H1.2 Exploration: Symmetry (continued)

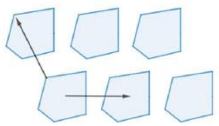
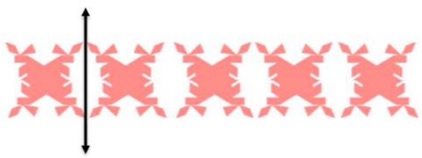
A ***tessellation*** is an arrangement of shapes closely fitted together in a repeated pattern without gaps or overlaps.

3. Which image is a tessellation?
   Image C is a tessellation.

Translational symmetry is when sliding a pattern is used to create symmetry.

Reflectional symmetry is when a pattern can be reflected to create symmetry.

Rotational symmetry is when a rotation can be used to create symmetry. The table shows examples of each type of symmetry.

Image	Symmetry
	Translational symmetry
	Reflectional symmetry
	Rotational symmetry

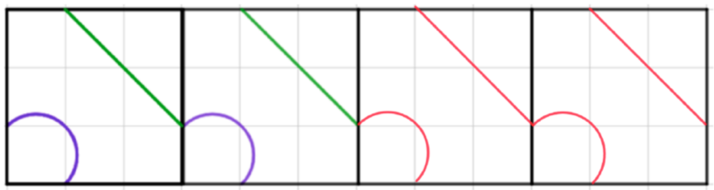
Consider implementing a "stop" after question 3 and check-in with student learning as a whole group.

1.H1.2 Exploration: Symmetry (continued)

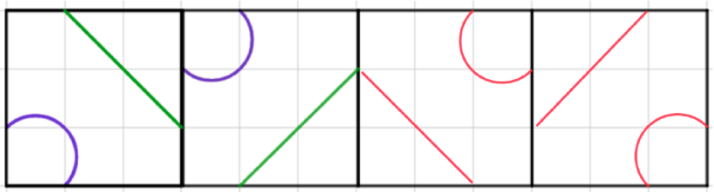
4. Micah commented to his partner that he thought the picture for reflectional symmetry could also be translational symmetry. Discuss this with your partner.

5. For each pattern, use the first two squares to complete the pattern.

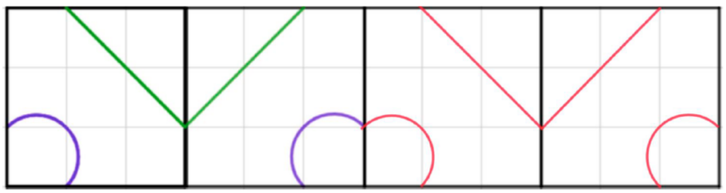
Pattern A



Pattern B



Pattern C



As you circulate consider engaging students in a discussion of why the image represents the symmetry listed. Ask students to include what should be looked for in determining the type of symmetry.

6. Work with your partner to determine the type of symmetry each pattern displays.

Pattern	Type of Symmetry
A	*Translational symmetry*
B	*Rotational symmetry*
C	*Reflectional symmetry*

This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

The order of rotational symmetry of a shape is the number of times it can be rotated around a full circle and still look the same.

7. Complete the chart for each polygon.

Polygon	Figure	Number of Lines of Symmetry	Order of Rotational Symmetry
Isosceles triangle		1	1
Equilateral triangle		3	3
Square		4	4
Kite		1	1
Regular Pentagon		5	5
Regular Hexagon		6	6

For a kinesthetic entry-point, provide a copy of the image that can be manipulated to determine the type of transformation prior to the students trying to draw the third and fourth images.

1.H1.3 Activity: Creating a Tessellation

Component Overview: *Students create a tessellation.*

1.H1.3 Activity: Creating a Tessellation

Image C is a tessellation that was created by MC Escher, a famous Dutch artist.

To create a tessellation, you will need a 3x5 index card and tape.

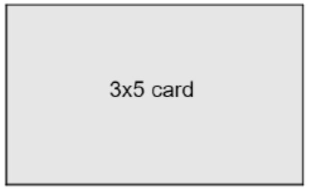
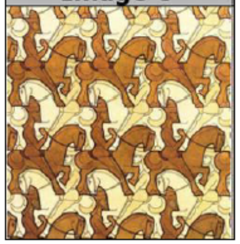


Image C

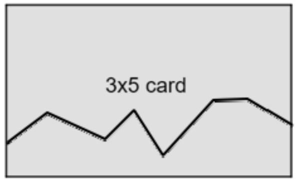


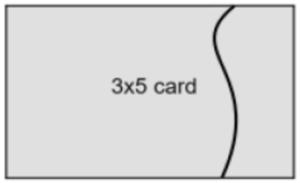
Colored pencils will allow students to add color. This activity can be started in class and completed at home. The teacher can display the completed art.

1. Create your own tessellation by choosing one of the following options. The more cuts you make, the more interesting your pattern will become.

Horizontal Cut
Draw a curve or pattern along a horizontal side of the card.

Vertical Cut
Draw a curve or pattern along a vertical side of the card.

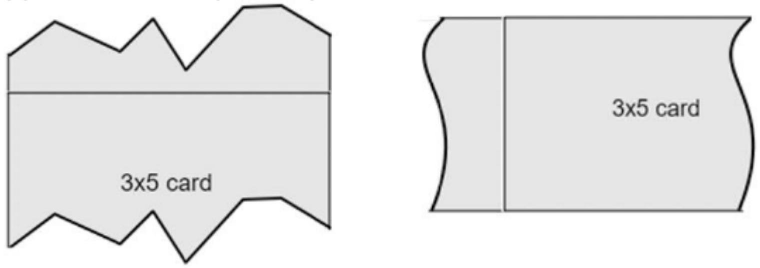




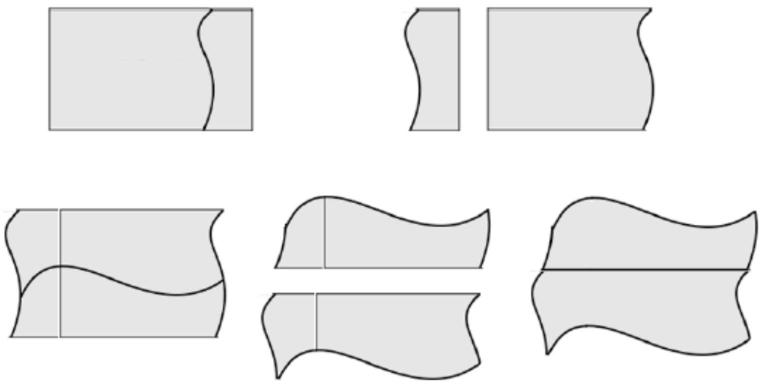
Consider collaborating with an art or graphic design instructor to see if tessellations are in their course guides. Engagement can be increased by giving students real-world meaning through cross-curricular instruction. Students can transfer content to multiple academic disciplines throughout this tessellation activity.

1.H1.3 Activity: Creating a Tessellation (continued)

Cut along the curve or pattern and shift the piece to the opposite side. Tape the piece on.



It is also possible to make multiple cuts as shown.



2. Use your card to create a tessellation by filling up a sheet of paper. Then, color your tessellation.

Students moving through the process of cutting and rejoining the note card can significantly engage students in a kinesthetic way. The addition of color can significantly improve the students' visual connection with the content.

MA.K12.MTR.5.1: Patterns and Structure
Students will be spending time focusing on relevant details in their pattern and creating plans to logically produce an overall image. This activity helps students recognize the patterns in the world around them and make connections to mathematical concepts.

1.H1.4 Wrap-Up: Cool Down

Component Overview: *Students identify the transformations that show symmetry for two real-world objects.*

1.H1.4 Wrap-Up: Cool Down

1. Symmetry is often found in nature. Describe the transformations that show the symmetry of each object.

Image	Object	Transformation
	Each petal	*Reflection and Rotation*
	Each arm of the starfish	*Reflection and Rotation*

Consider having the students take pictures of objects with symmetry and to share them with you for a class slideshow.